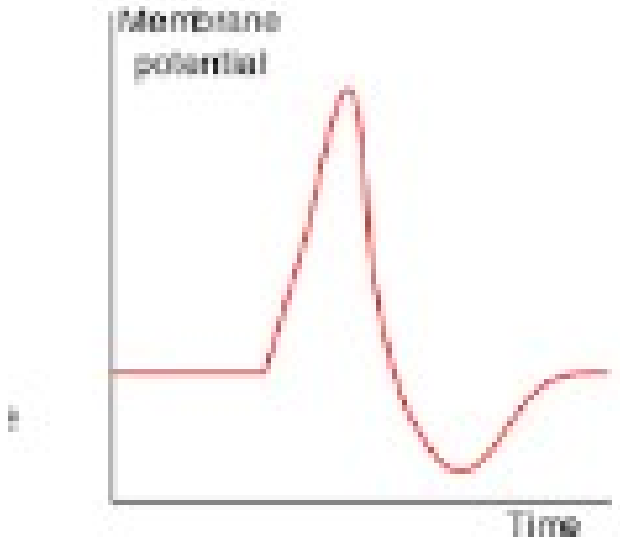
Nerve Impulses – Action Potentials



Neurons Communicate Quickly

Nerve Impulses

- A nerve impulse message that travels along a nerve fibre (neuron)
- Electrochemical change that travels along neuron cell membrane:
  - Changes in ion concentrations across the cell membrane

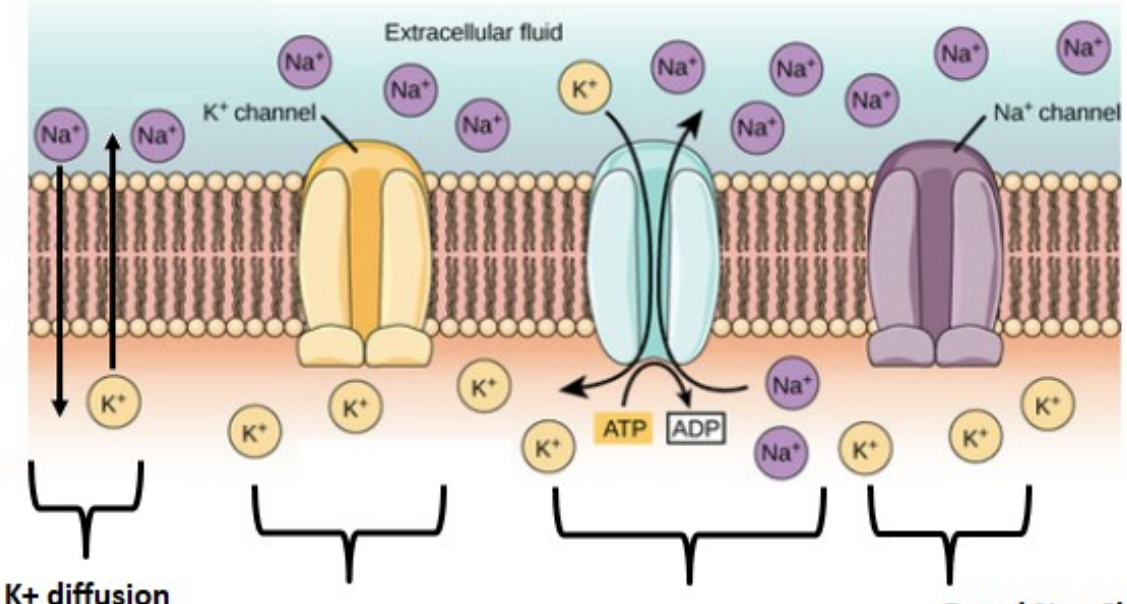
    lead to
  - Changes in electrical voltage called ***Action Potentials (AP)***

    which
  - Travel as a wave along the axon

- Can vary in speed of transmission:
  - Myelinated vs unmyelinated
  - Diameter of fibre: larger diameter = faster transmission
- Fast response to stimulation

Learning Aim: Define the term "Action Potential (AP)"

Parts of the Neuron Cell Membrane



Outside the neuron

Inside the neuron

Na+ and K+ diffusion across cell membrane

happens passively, all the time, which is why we need the Na+/K+ pump in neuron membranes

Gated K+ Channel

allows extra K+ across membrane when open.

Na+/K+ Pump
active transport: uses ATP

Pumps 3 sodium out of cell and pumps 2 potassium in.

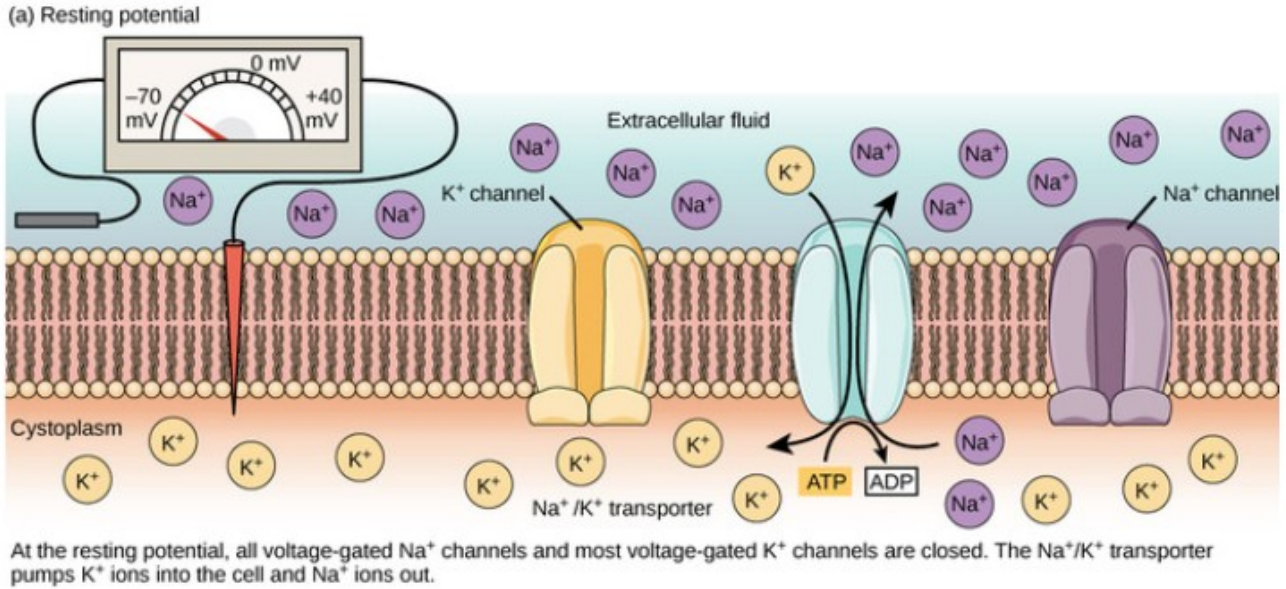
Happens all the time

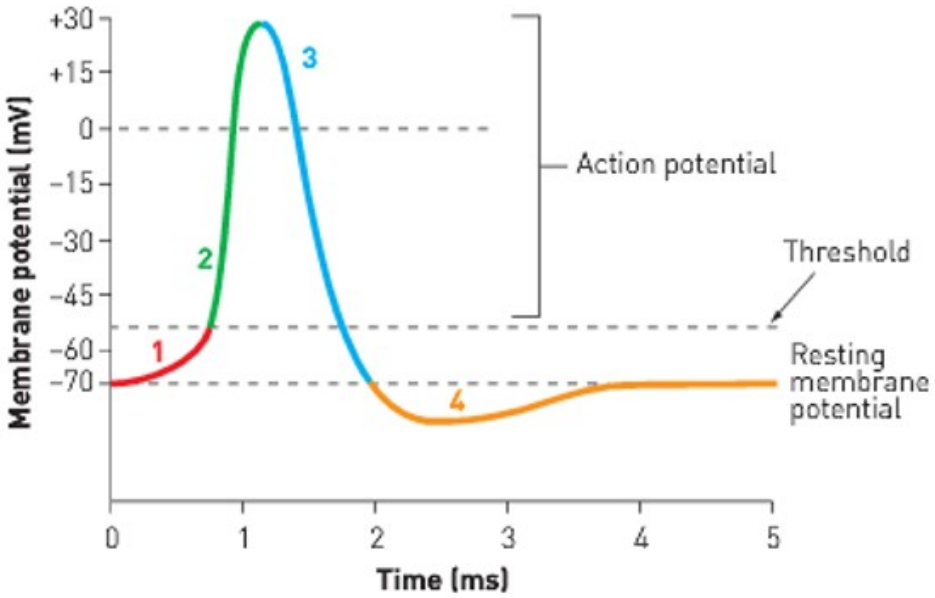
When membrane is at rest, it keeps the outside *relatively* positive, compared to the inside.

Gated Na+ Channel
allows extra Na+ to cross the membrane when open.

Learning Aim: Describe features of the neuron cell membrane that allow ion movement.

Resting Membrane Potential (RMP): Before Neuron is stimulated

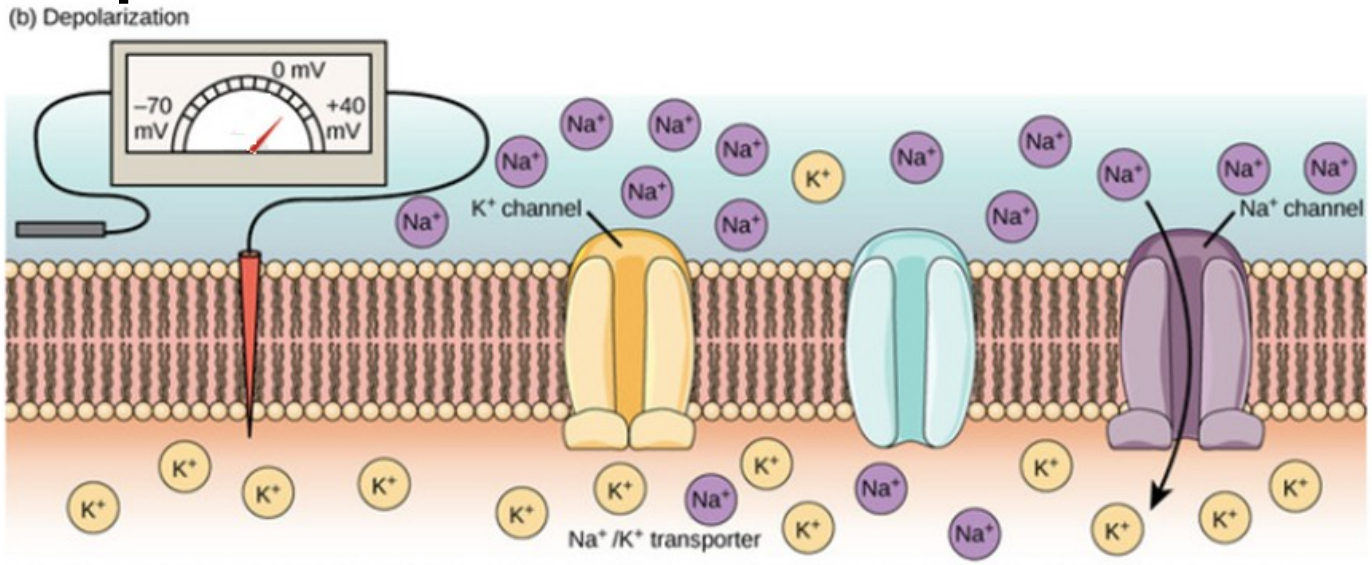




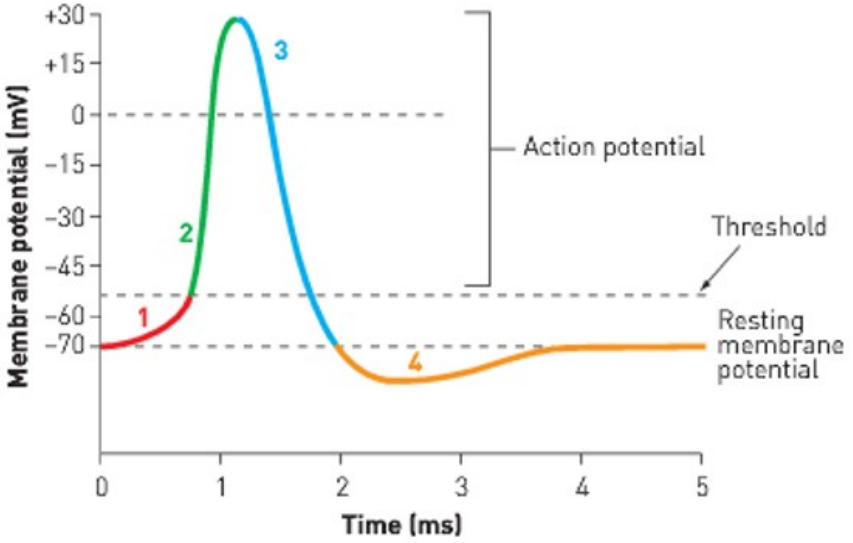
- Inside of neuron relatively negatively charged, compared to outside, because of Na+/K+ pump
- Gated K+ and Na+ channels closed

Learning Aim: Describe features of the neurone cell membrane such as relative position and concentration of ions, membrane potential and channel activity during Resting Membrane Potential (RMP)

Depolarisation: when neuron is stimulated



In response to a depolarization, some Na⁺ channels open, allowing Na⁺ ions to enter the cell. The membrane starts to depolarize (the charge across the membrane lessens). If the threshold of excitation is reached, all the Na⁺ channels open.



i: Local stimulation causes local changes in electrical potential by opening a few Na+ channels. (1)

ii: Threshold of excitation is reached -----

iii: Many gated Na+ Channels open

iv: Na+ floods across membrane (2)

v: Inside of cell is now relatively positive.

Note:

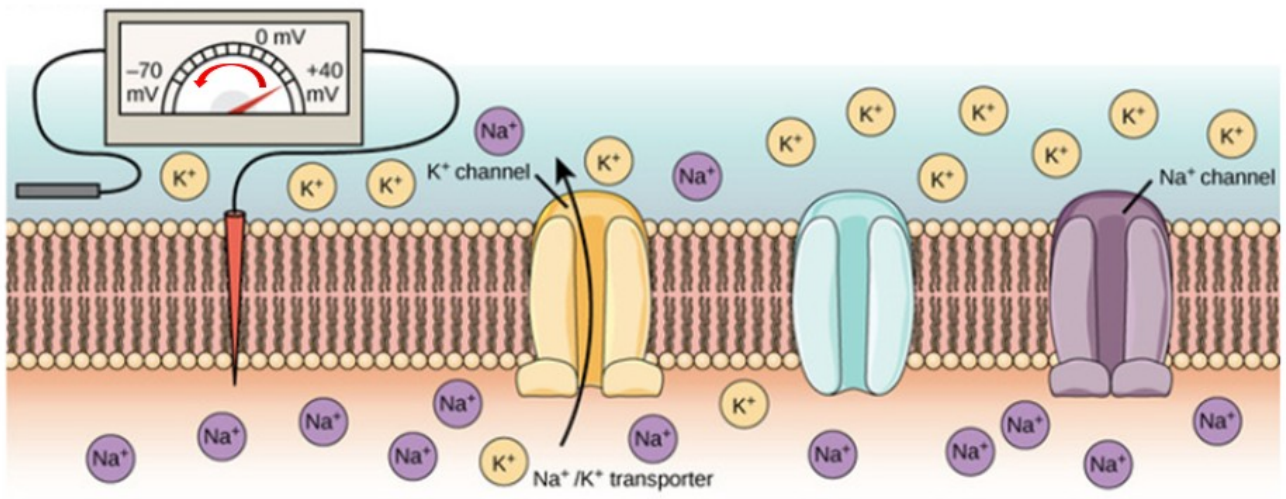
If the threshold of -55mV is not reached, an Action Potential (AP) will not be triggered.

*For each individual neuron, it's an **"all or nothing"** response.*

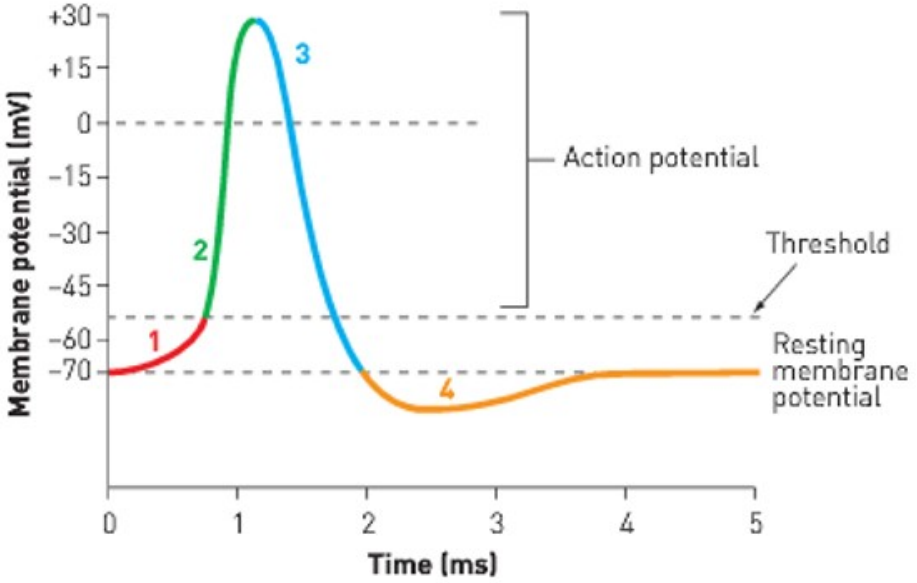
So, why do we feel it more if we drop a hammer on our foot compared to a slice of bread?

Learning Aim: Describe features of the neurone cell membrane such as relative position and concentration of ions, membrane potential and channel activity during Depolarisation

Repolarisation



At the peak action potential, Na^+ channels close while K^+ channels open. K^+ leaves the cell, and the membrane eventually becomes hyperpolarized.



At peak of action potential:

i: Gated K+ Channels open, and Na+ channels close

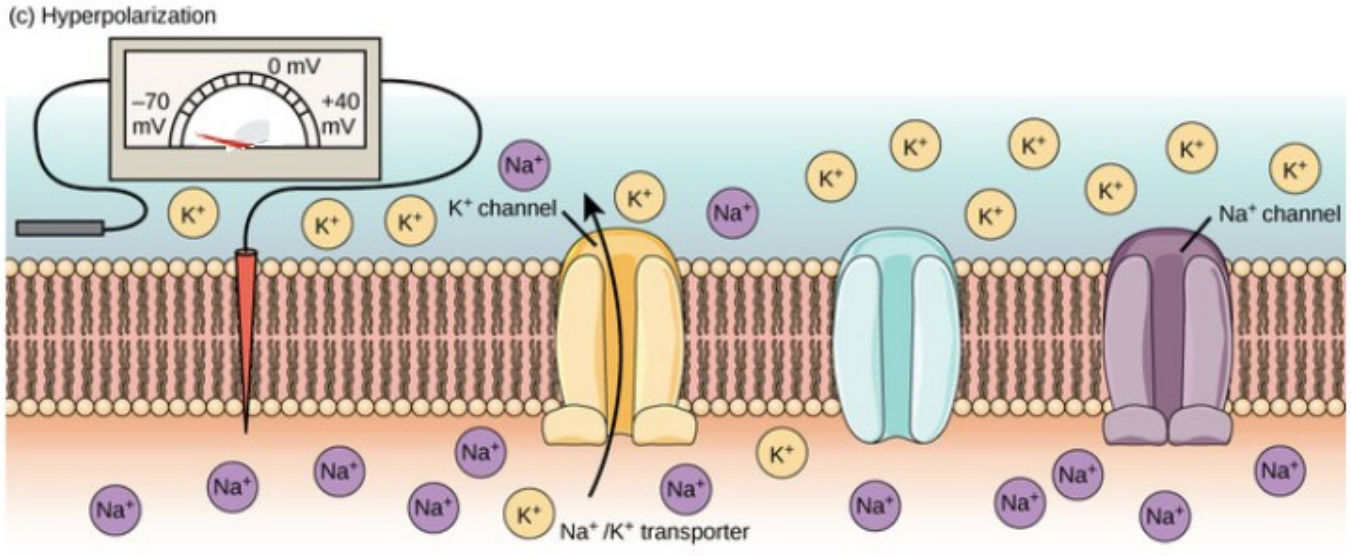
ii: K+ floods out across membrane

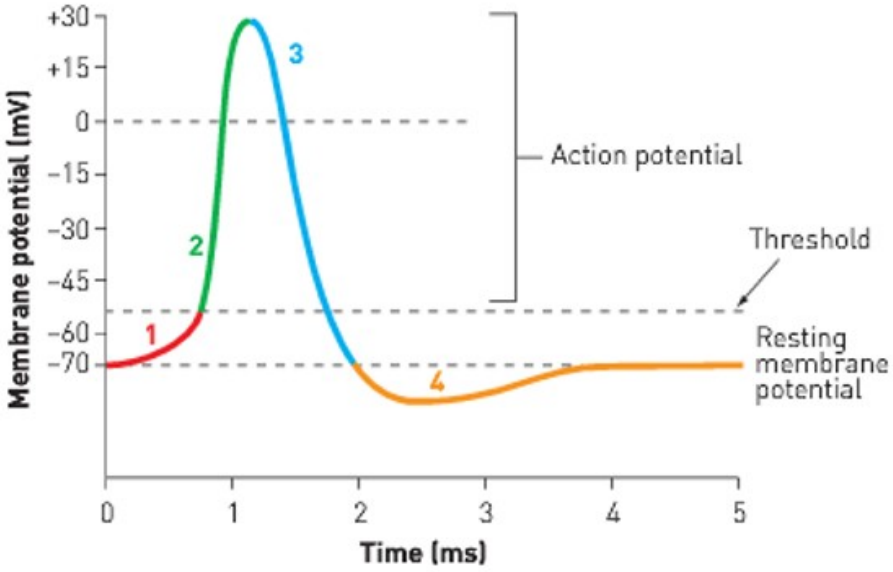
iii: Charge begins to fall, back to -70mV

(3)

Learning Aim: Describe features of the neurone cell membrane such as relative position and concentration of ions, membrane potential and channel activity during Repolarisation

Hyperpolarisation



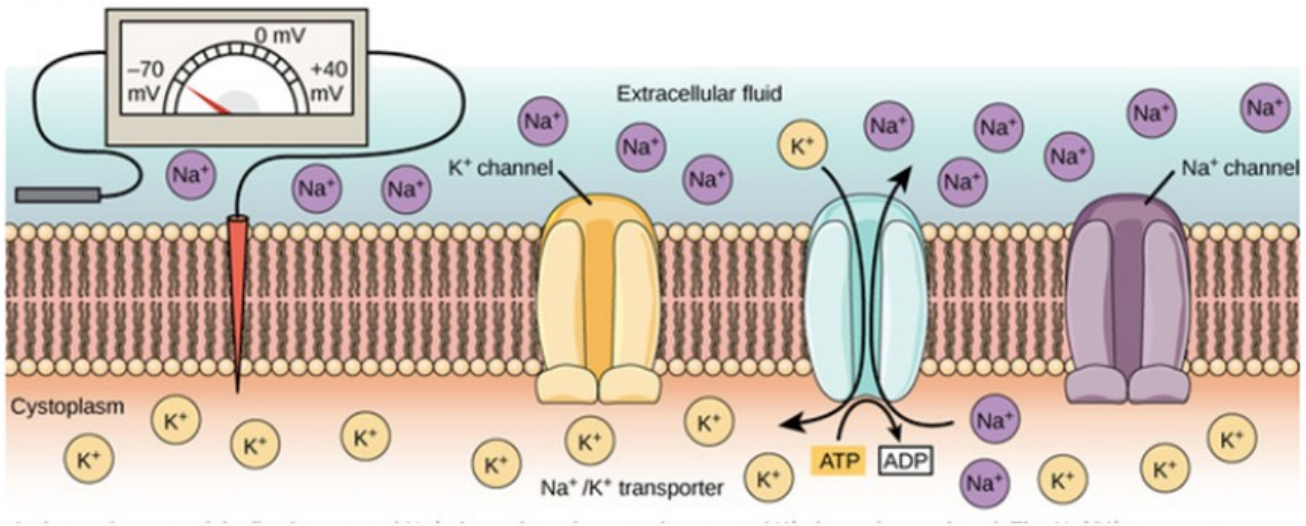


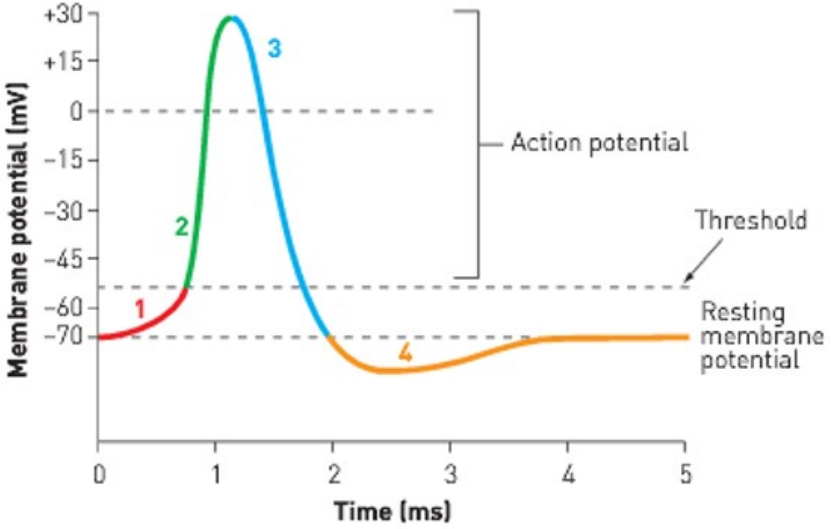
K+ channels remain open, and eventually the inside of the cell becomes even more negative than it was at rest. This is called ***hyperpolarisation.*** (4)

Note that now the K+ is on the outside, so it isn't back to rest – the Na+/K+ pump needs to work to pump Na+ out and K+ in to regain resting potential.

Learning Aim: Describe features of the neurone cell membrane such as relative position and concentration of ions, membrane potential and channel activity during Hyperpolarisation

Return to Resting Membrane Potential (RMP)

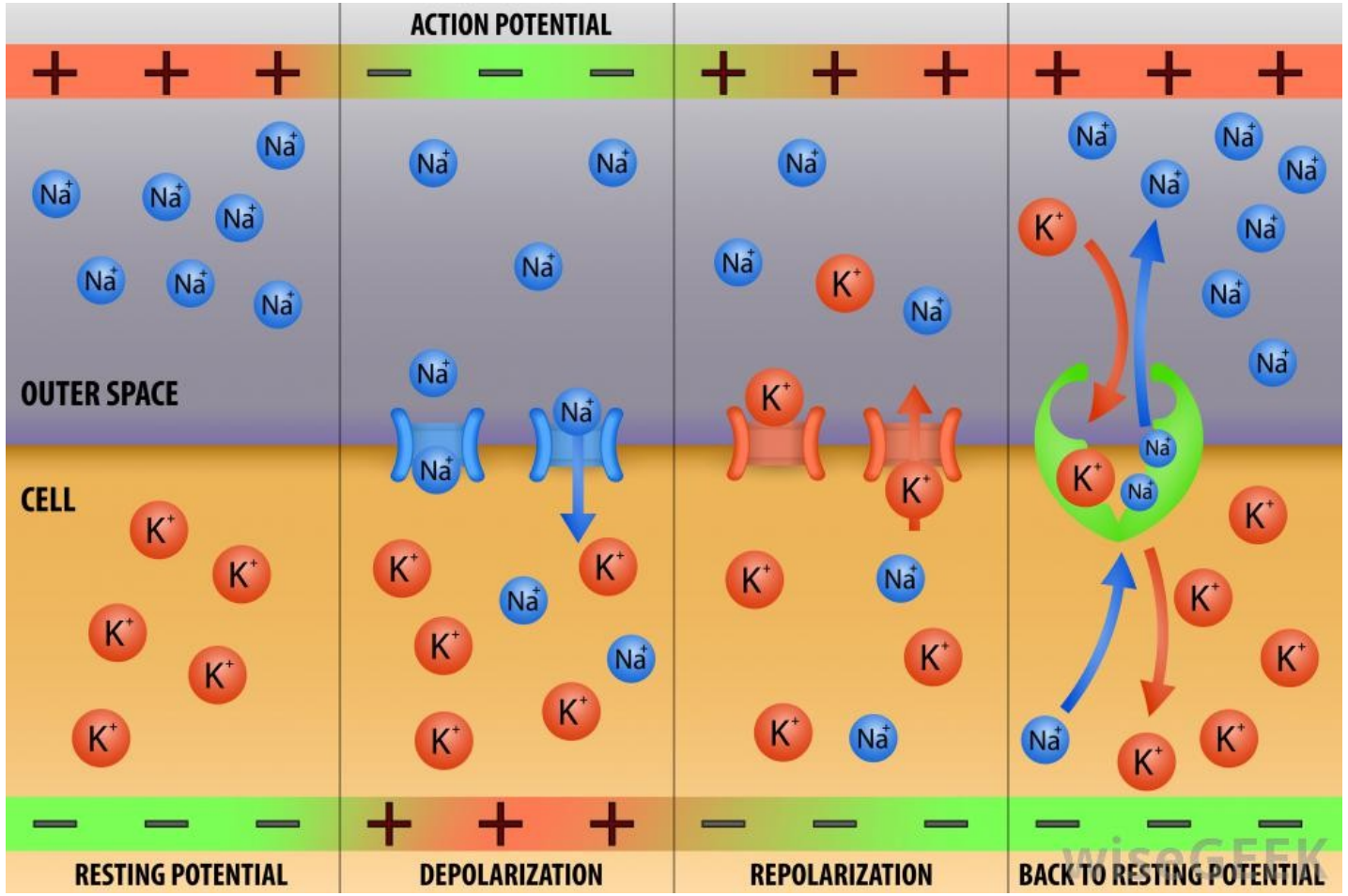


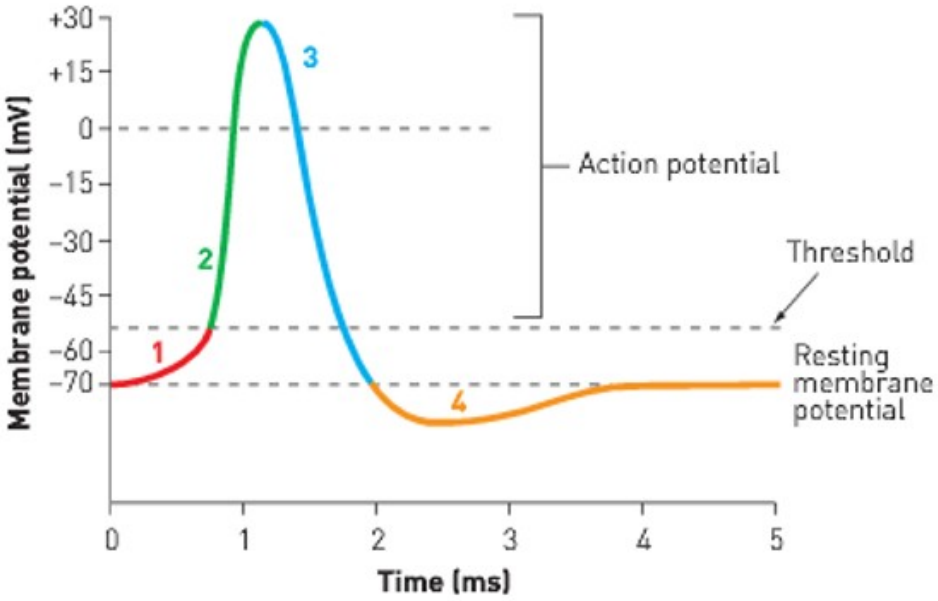


- Gated K+ channels close.
- The Na+/K+ pump continues to work, eventually returning the membrane potential to resting status.

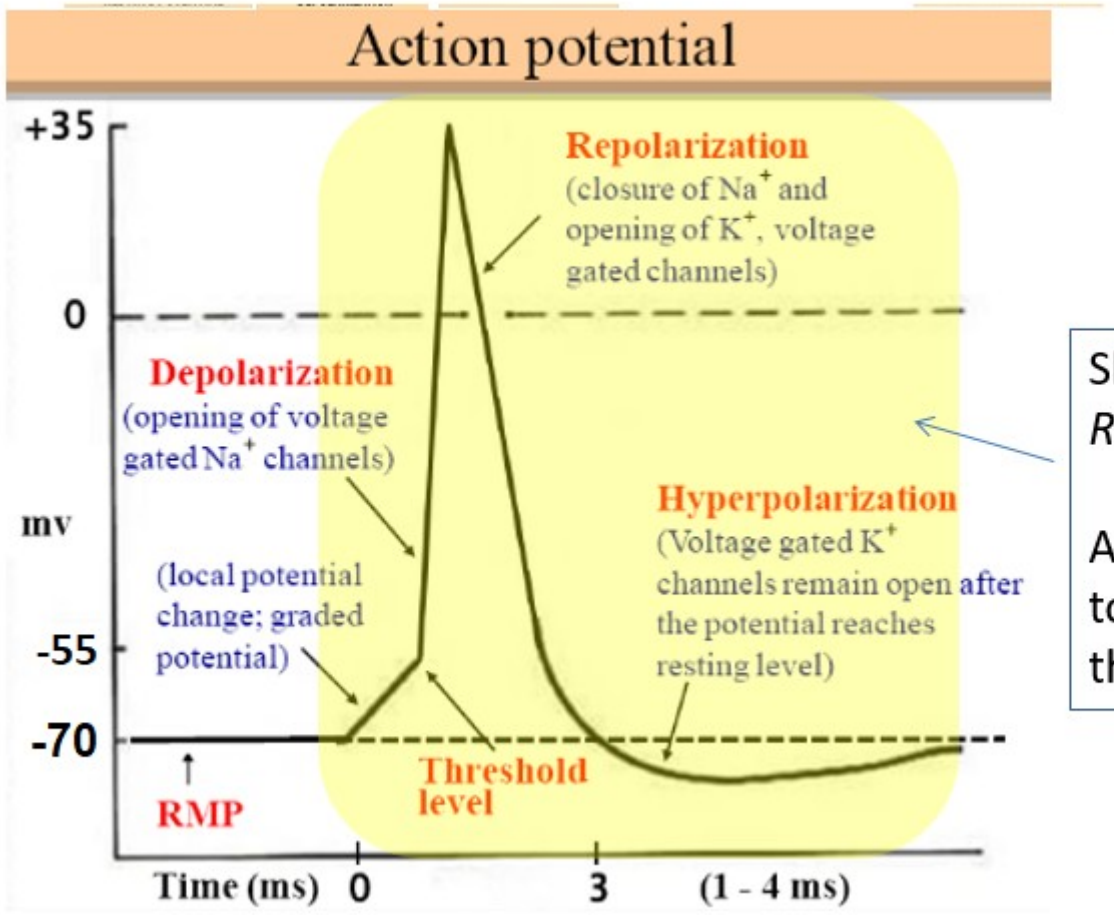
Learning Aim: Describe features of the neurone cell membrane such as relative position and concentration of ions, membrane potential and channel activity during return to RMP

All the steps in one





The Refractory Period



Shaded area shows *Refractory Period:*

Axon won't respond to further stimulus in this time

The *refractory period* is the reason an AP travels **only in one direction** along the membrane.

(dendrites -> cell body -> axon)

Due to the concentration of ions on either side of the membrane and the membrane potential at this time, the section of membrane that has just had a wave of depolarisation can't have another one, so the stimulus only moves to a section that has not already had an action potential.

Learning Aim: Define the term "refractory period" and explain its importance in AP transmission

Propagation of AP along the neuron

- An action potential in one section of membrane triggers a new one I the section of membrane immediately next to it, and so on down the neuron.

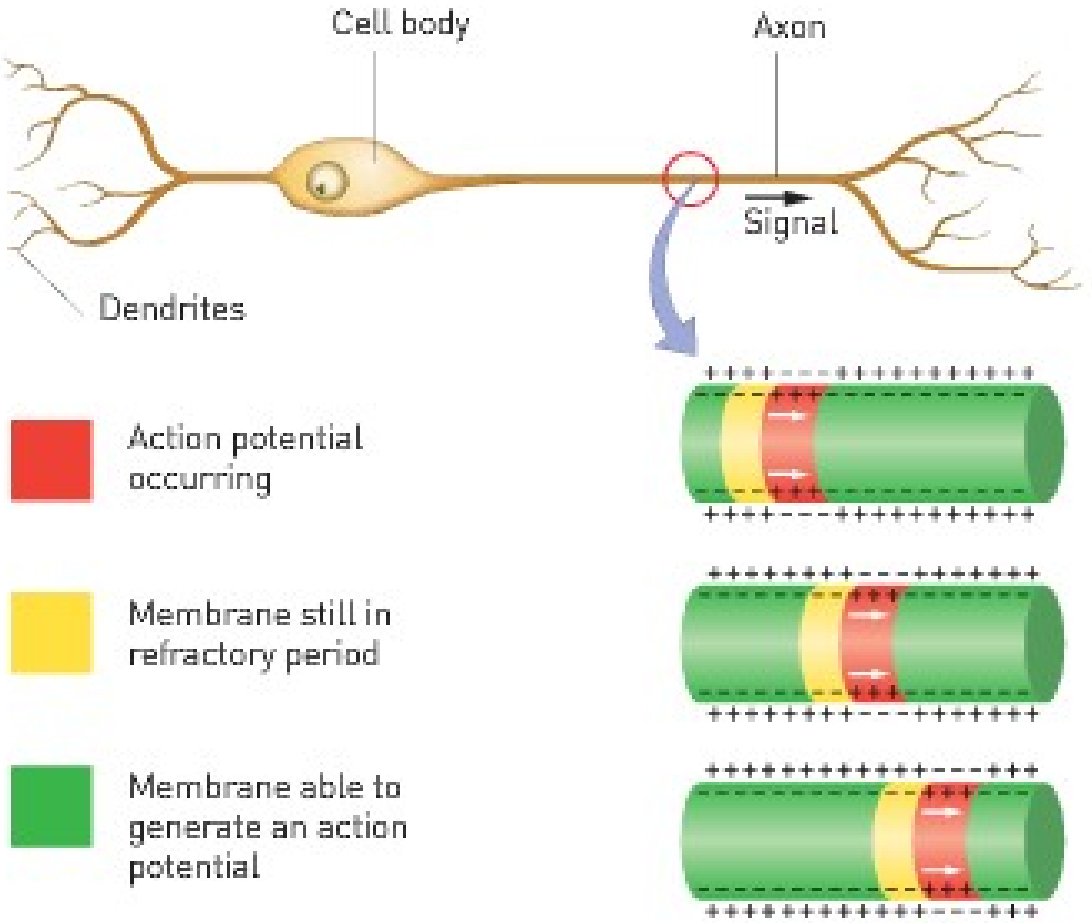


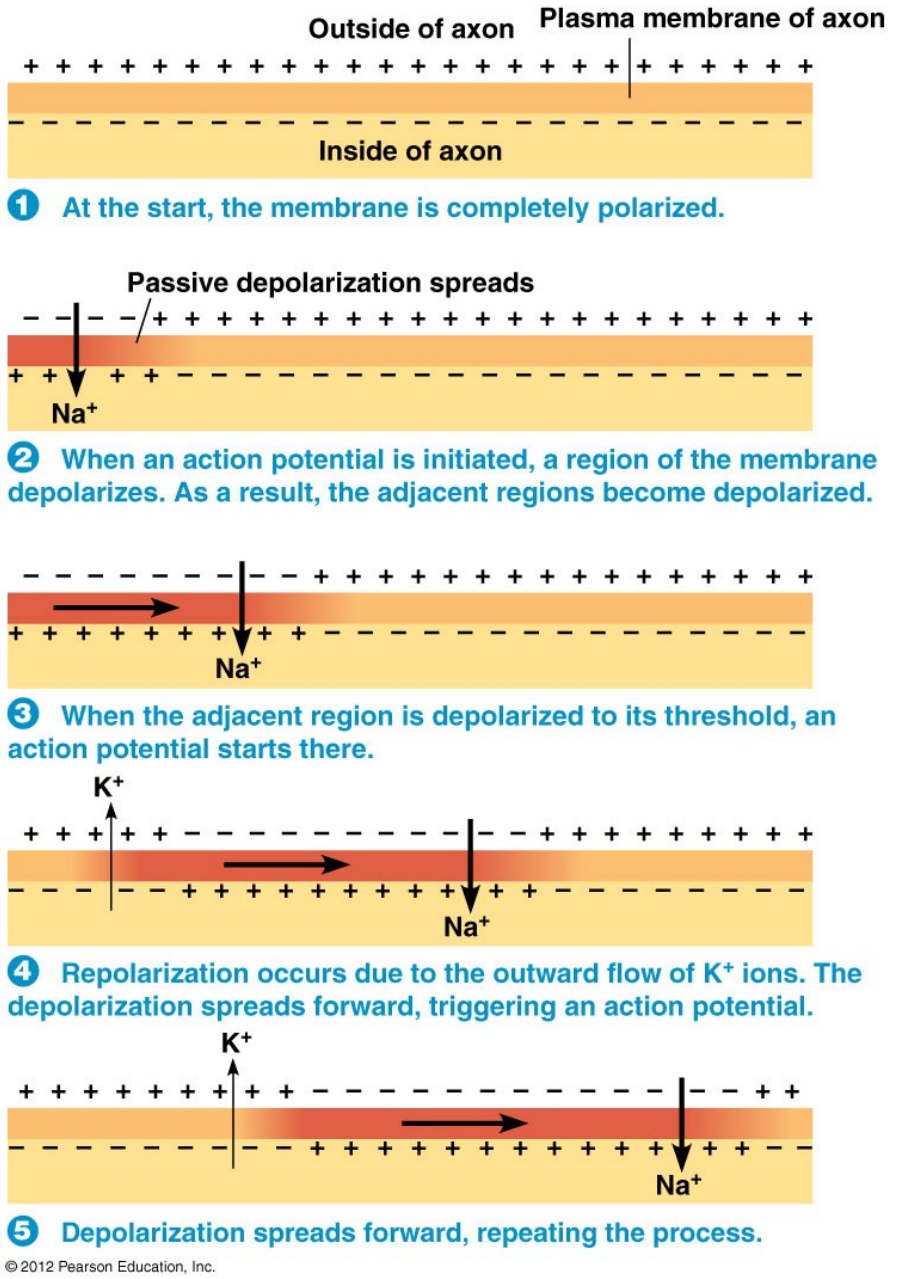
Figure 3.11 Transmission of a nerve impulse along an unmyelinated fibre. Successive action potentials are generated along the membrane of the nerve fibre.

Learning Aim: Explain how the AP propagates/transmits along the neuron membrane

How an Action Potential Moves along an axon



Depolarisation in one part of the membrane triggers depolarisation in the next, just as when a domino falls, it pushes the next one over.



Learning Aim: Explain how the AP propagates/transmits along the neuron membrane

Action potentials along myelinated axons: *Saltatory Conduction*.

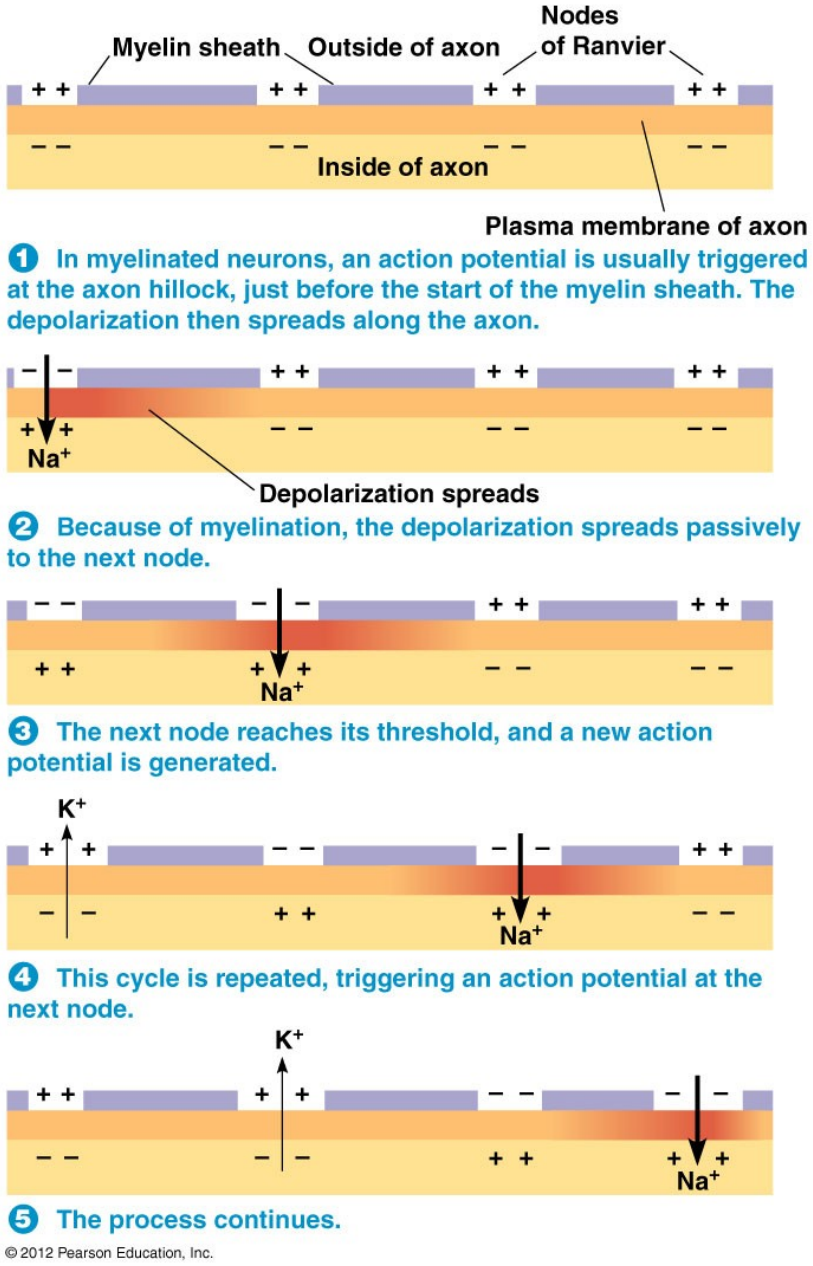
Saltatory Conduction occurs in myelinated axons.

The myelin sheath covers sections of the cell membrane so that depolarisation can only occur at the nodes of Ranvier.

This means that the initial influx of sodium ions quickly diffuses to the next node, rather than triggering depolarisation in the membrane immediately next to it.

The sodium ions quickly and passively diffuse down the axon until the next node.

This makes the depolarisation jump from node to node, causing quicker transmission.



Learning Aim: Describe what occurs during saltatory conduction and its effect on AP transmission

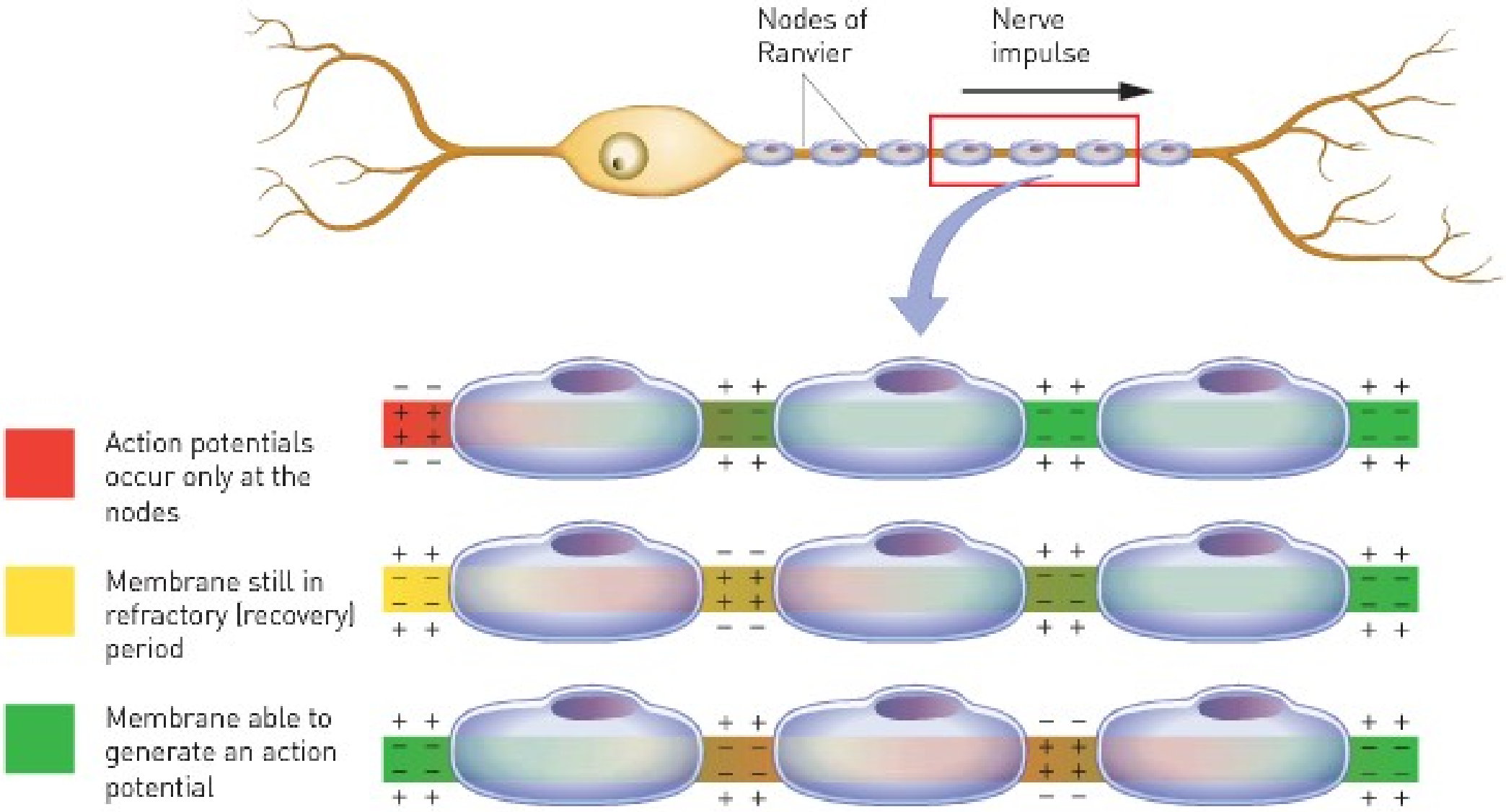


Figure 3.12 Saltatory conduction along a myelinated fibre. The nerve impulse jumps from node to node.

Learning Aim: Describe what occurs during saltatory conduction and its effect on AP transmission

An extra diagram showing the same things about the AP cycle but in a different format….

